INTRODUCTION

MICROPROCESSORS & INTERFACING (CS F241)

Dr. Manideepa Mukherjee Bhakat (IC)
Department of CS & IS

Course Details

Course Title: Microprocessors & Interfacing

Course No.: CS F241

Instructor-in-charge: Dr. Manideepa Mukherjee

Instructor: Dr. Gargi Alavani - Prabhu

Class Time: **M W** 12:00 -12:50 **TH** 16.00-16.50

Tutorial: **M** 8.00- 9.00

Lab: **T** 11.00 -13.00

Announcement:
– All notices concerning this course will be updated on the quanta course page or communicated in class.

Reference Material

BITS Pilani
K K Birla Goa Campus

Textbook:

T1	Barry B Brey, The Intel Microprocessors, Pearson, Eight Ed. 2009

Reference Books:

R1	Douglas V Hall, Microprocessor and Interfacing, TMH, Second Edition
R2	80x86 Manuals
R3	Robert L. Britton, MIPS Assembly Language Programming

Evaluation Components

Quiz – 10% (Closed Book)
Quiz I: 03/02/2024
Quiz II: 24/02/2024
Quiz III: 07/04/2024

Lab – 30 %
Mid Semester – 25% (*Partially Closed Book)
End Semester – 35% (*Partially Closed Book)

Make-up Policy:
- Prior permission is needed for genuine cases.
- Health reasons will need a certificate from Medical Centre.
- Granting make-up for midsemester is the sole discretion of the IC.

Plan for Lab work

Lab Number	Date	Details
Lab 1	20/01/2026	
Lab 2	27/01/2026	
Lab 3	03/02/2026	Quiz-I
Lab 4	10/02/2026	
Lab 5	17/02/2026	Evaluative
Lab 6	24/02/2026	Quiz-II
Lab 7	17/03/2026	Evaluative
Lab 8	24/03/2026	
Lab 9	07/04/2026	Quiz-III
Lab 10	21/04/2026	Evaluative

ANY QUESTIONS RELATED TO HANDOUT?

Why learn about Microprocessors?

Why learn about Microprocessors?

“Every AI model, cloud service, and smartphone app ultimately runs on *this*…. a MICROPROCESSOR. “

So why do most people want to skip understanding it?

Why learn about Microprocessors?

Have you ever thought about?

- Why your phone battery drains?
- Why GPUs are fast but power-hungry?
- Why embedded systems still dominate the real world?
- Why AI on the edge needs low-power processors?

This is exactly where modern processor design and interfacing matter!

What is Microprocessors & Interfacing?

Microprocessor & Interfacing is the study of how a processor executes instructions and how it communicates with external devices to interact with the real world.

A microprocessor is the “brain” of a computing system that executes instructions and processes data.

Interfacing is the technique of connecting the microprocessor to external devices and enabling communication between them.

Let’s Clear the Fear First

- This course is **NOT** about memorizing pin diagrams or opcodes.

- It’s about understanding how software meets hardware.

> - **Assembly *looks* scary**
> - **Interfacing *seems* messy**

innovate achieve lead

Let’s Clear the Fear First

- This course is **NOT** about memorizing pin diagrams or opcodes.

- It’s about understanding how software meets hardware.

- **As[redacted]ary**
- **Interfa[redacted]messy**

- **Once you understand the core idea, everything starts to make sense.**
- **You start *reasoning*, not memorizing**

Course Outline

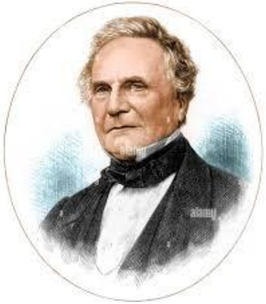
- Introduction & history of Microprocessors
- Architecture
- Assembly Programming
- Memory Interface
- I/O Interface
- Interrupts
- DMA Controller
- Bus Interface
- Advanced Processors : 80286, MIPS

A Historical Background

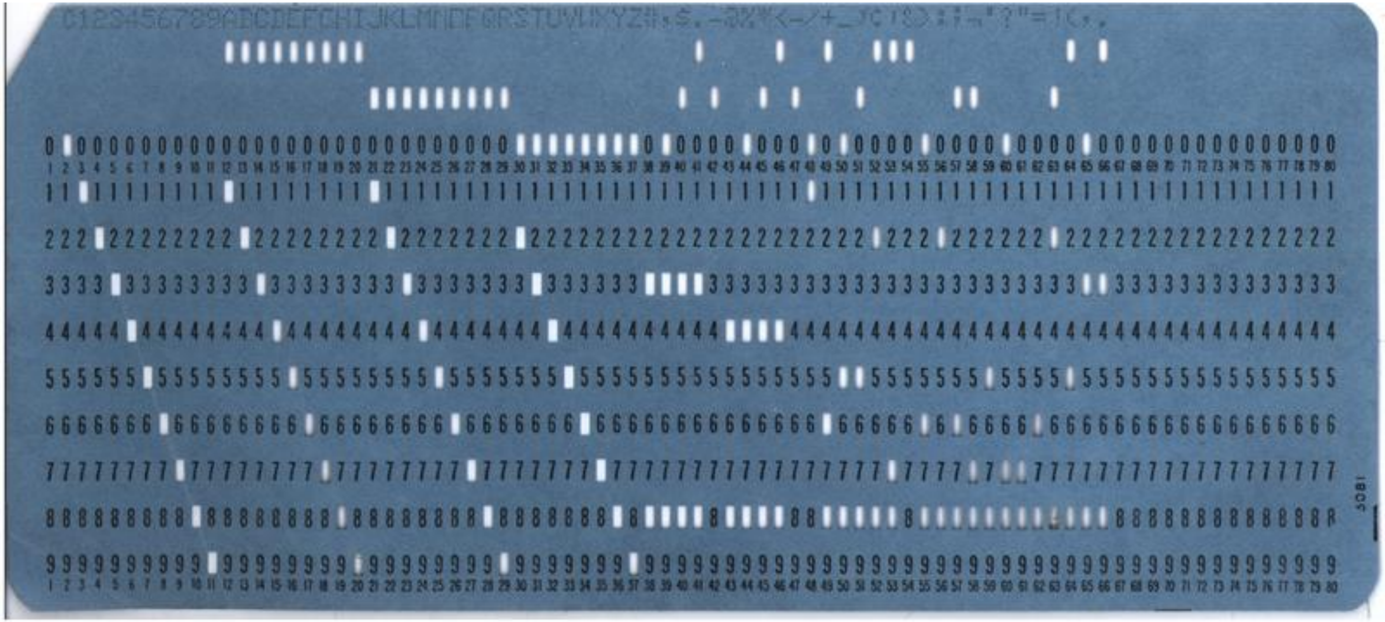
- Blaise Pascal (1642) invented a calculator constructed of gears and wheels.
  - Each gear contained 10 teeth when moved one complete revolution, a second gear advances one place.

A Historical Background : Mechanical-> Electrical

- One early pioneer of mechanical computing machinery was **Charles Babbage**.
  - aided by Ada Byron, Countess of Lovelace
- Commissioned in 1823 by Royal Astronomical Society to
- build programmable calculating machine.
  - to generate Royal Navy navigational tables
- 1800s saw advent of the electric motor.
  - conceived by **Michael Faraday**
- Introduced by Bomar Corporation the **Bomar Brain**, was a handheld electronic calculator.
  - first appeared in early 1970s
- In 1889, **Herman Hollerith** developed **the punched card for storing data**.



A Historical Background



A Historical Background

- Mechanical-electric machines dominated information processing world until 1941.
  - construction of first electronic calculating machine
- German inventor **Konrad Zuse**, invented the first modern electromechanical computer.
- His **Z3** calculating computer probably invented for aircraft and missile design.
  - during World War II for the German war effort
- Z3 a relay logic machine clocked at 5.33 Hz.
  - far slower than latest multiple GHz microprocessors
  - Lacked stored program concept

A Historical Background

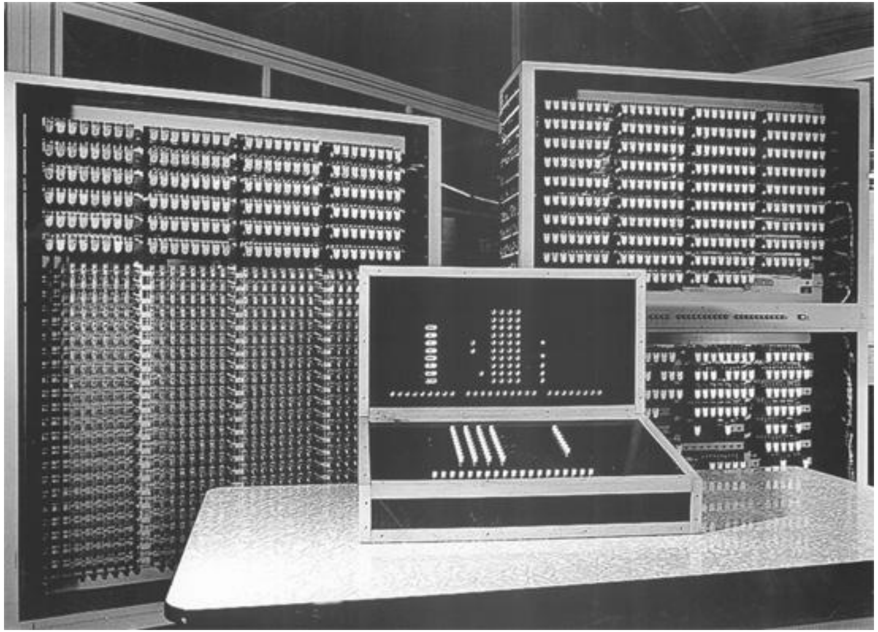
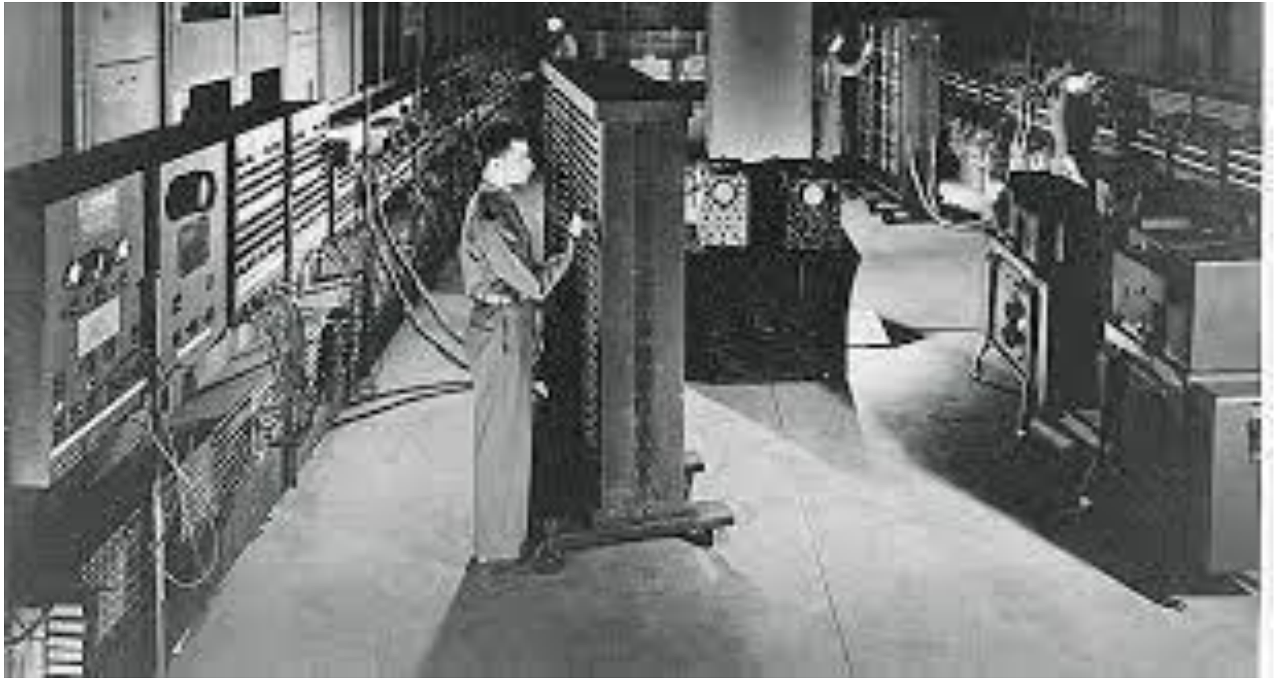


Figure. The Z3 computer developed by Konrad Zuse uses a 5.33 hertz clocking frequency.

A Historical Background

- System invented by **Alan Turing**.
  - used vacuum tubes,
- Turing called his machine **Colossus.**
- A fixed-program computer system
  - today often called a **special-purpose computer**
- **Electronic Numerical Integrator and Calculator (ENIAC)**
- First general-purpose, programmable electronic computer system developed in1946.
  - at University of Pennsylvania
- a huge machine.
  - over 17,000 vacuum tubes; 500 miles of wires
  - weighed over 30 tons
  - about 100,000 operations per second

A Historical Background



ENIAC Machine

A Historical Background

- **Electronic Numerical Integrator and Calculator (ENIAC)**
- Programmed by rewiring its circuits.
  - process took many workers several days
  - workers changed electrical connections on plug-boards like early telephone switchboards
- Required frequent maintenance.
  - vacuum tube service life is a problem
- December 23, 1947, **John Bardeen, William Shockley, and Walter Brattain** develop the transistor at Bell Labs.
- Followed by 1958 invention of the integrated circuit (IC) by **Jack Kilby** of Texas Instruments.

Vaccum Tube

Transistor

The Microprocessor

- Called the CPU (**central processing unit**).
- The controlling element in a computer system.
- Controls memory and I/O through connections called buses.
    - buses select an I/O or memory device, transfer data between I/O devices or memory and the microprocessor, control I/O and memory systems
- Memory and I/O controlled via instructions stored in memory, executed by the microprocessor.

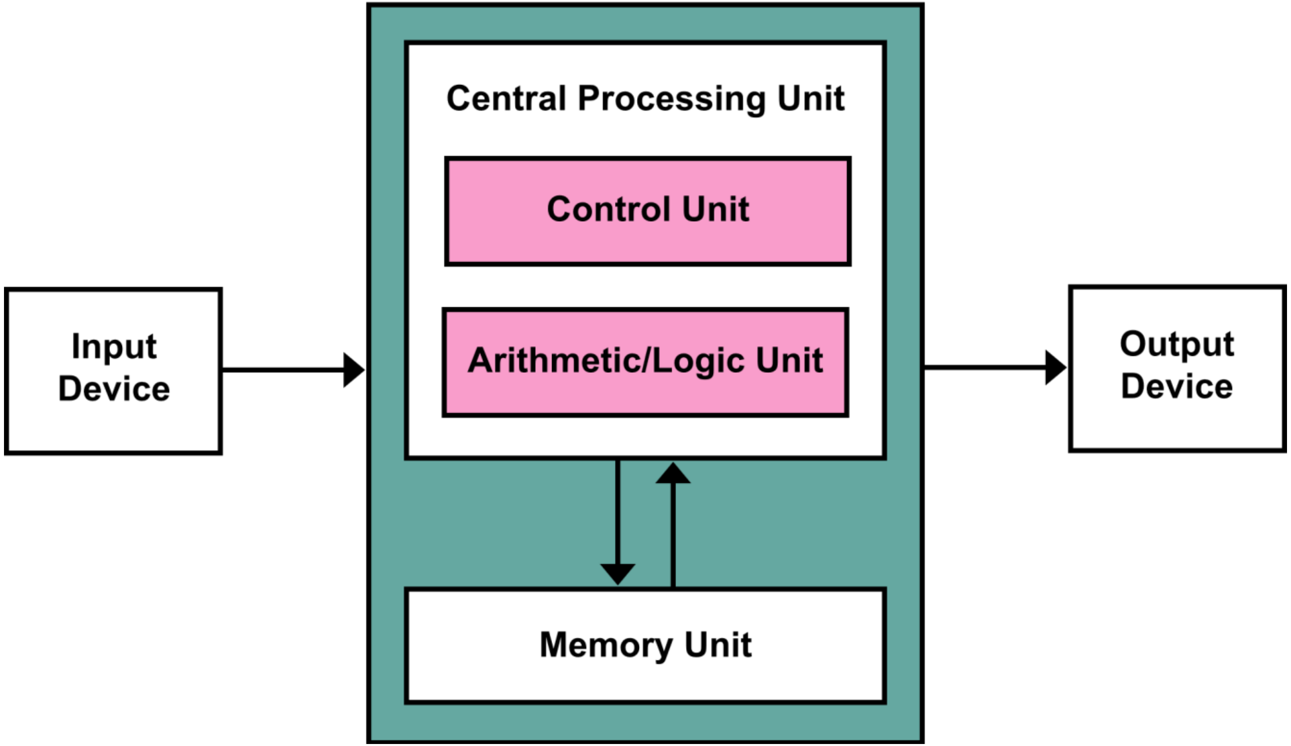
Von Neumann Machines (1945)

- More efficient than rewiring a machine to program it.

- Mathematician **John von Neumann** first modern person to develop a system to accept instructions and store them in memory.

- Computers are often called **von Neumann machines** in his honor.

Von Neumann Machines



A Historical Background

- First microprocessor developed at Intel Corporation in **1971**.
- Intel engineers Federico Faggin, Ted Hoff, and Stan Mazor developed the 4004 microprocessor.
- U.S. Patent 3,821,715
- Device started the microprocessor revolution continued today at an ever-accelerating pace.
- The first, **machine language**, was constructed of ones and zeros using binary codes.
  - stored in the computer memory system as groups of instructions called a program

A Historical Background

- Once systems such as UNIVAC(**Universal Automatic Computer I**) became available in early 1950s, **assembly language** was used to simplify entering binary code.

- Assembler allows programmer to use mnemonic codes…
  - such as ADD for addition

- In place of a binary number.
  - such as 0100 0111

A Historical Background

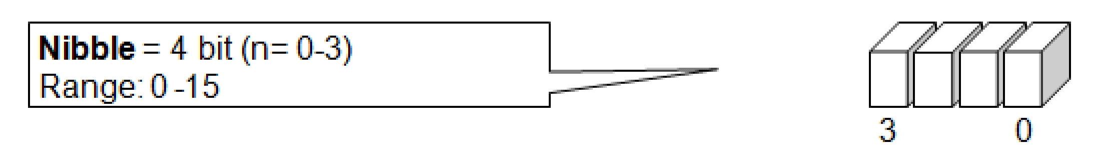
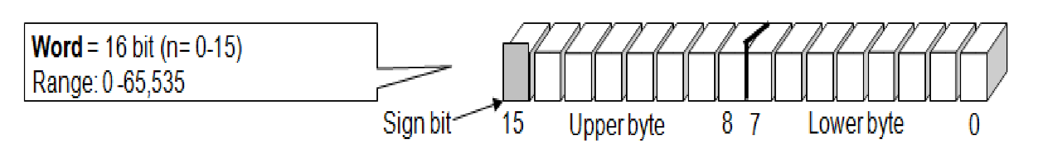
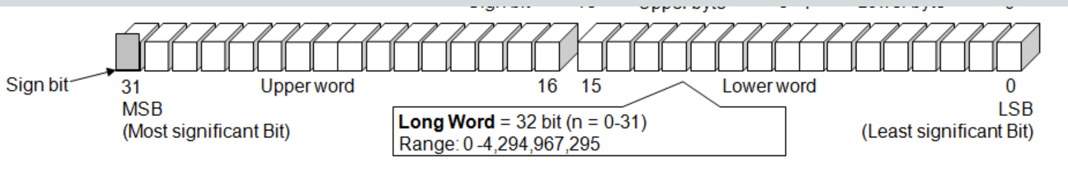
- 1957 Grace Hopper developed first high-level programming language called **FLOWMATIC.**
  - computers became easier to program
- In same year, IBM developed FORTRAN **FORmula TRANslator**) for its systems.
  - Allowed programmers to develop programs that used formulas to solve mathematical problems.
- FORTRAN is still used by some scientists for computer programming.
  - Similar language, ALGOL (**ALGOrithmic Language**) introduced about a year later

The Microprocessor Age

- World's first microprocessor the **Intel 4004**.
- Single Chip (2300 Transistors)
- A 4-bit microprocessor-programmable controller on a chip.
- Addressed 4096, 4-bit-wide memory locations.
  - a **bit** is a binary digit with a value of one or zero
  - 4-bit-wide memory location often called a **nibble**
- The 4004 instruction set contained 45 instructions.
- Executed 50KIPs << 100,000 IPs (30 Ton ENIAC)

Data Size

Data size in a microprocessor, referred to as the "word size" or "data bus width"

Nibble	**4 bit**	
Byte	8 bit	
Word	16 bit	
Long word	32 bit	

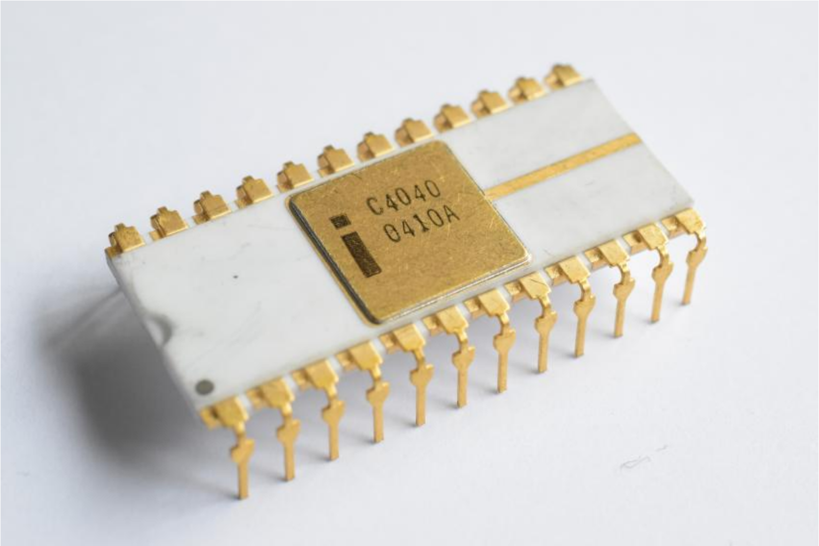
The Microprocessor Age

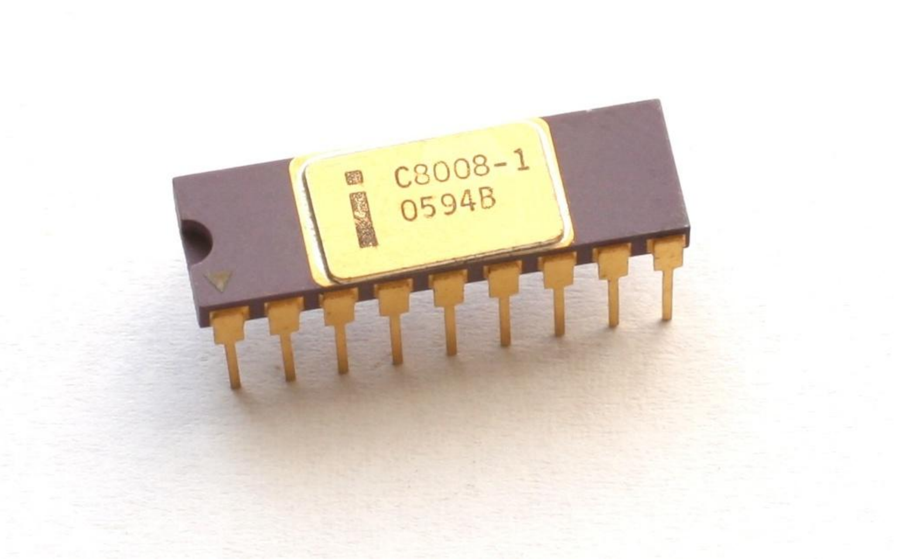
- Main problems with early microprocessor were speed, word width, and memory size.
- Evolution of 4-bit microprocessor ended when Intel released the 4040, an updated 4004.
  - operated at a higher speed; lacked improvements in word width and memory size
- Texas Instruments and others also produced 4-bit microprocessors.
  - still survives in low-end applications such as microwave ovens and small control systems
  - Calculators still based on 4-bit BCD (**binary-coded decimal**) codes

The Microprocessor Age

- Intel released 8008 in 1971.
    - extended 8-bit version of 4004 microprocessor
- Addressed expanded memory of 16K bytes.
    - A **byte** is generally an 8-bit-wide binary number and a **K** is 1024.
    - memory size often specified in K bytes
- Contained additional instructions, **48 total.**
- Provided opportunity for application in more advanced systems.
    - engineers developed demanding uses for 8008

Intel 4040, 8008





The Microprocessor Age

- Intel introduced 8080 microprocessor in 1973.
  - first of the modem 8-bit microprocessors
- Motorola Corporation introduced MC6800 microprocessor about six months later.
- 8080 addressed four times more memory.
  - 64K bytes vs l6K bytes for 8008
- Executed additional instructions; 10x faster.
  - addition taking 20 µs on an 8008-based system required only 2.0 µs on an 8080-based system
- TTL (transistor-transistor logic) compatible.
  - the 8008 was not directly compatible
- Interfacing made easier and less expensive.

The Modern Microprocessor

- In 1977 Intel Corporation introduced an updated version of the 8080—the 8085.

- Last 8-bit, general-purpose microprocessor developed by Intel.
- Slightly more advanced than 8080; executed software at an even higher speed.
    - 769,230 instructions per second vs 500,000 per second on the 8080).

The Modern Microprocessor

- In 1978 Intel released the 8086; a year or so later, it released the 8088.
- Both devices are 16-bit microprocessors.
    - executed instructions in as little as 400 ns (2.5 **millions of instructions per second**)
    - major improvement over execution speed of 8085
- 8086 & 8088 addressed 1M byte of memory.
    - 16 times more memory than the 8085
    - **1M-byte memory** contains 1024K byte-sized memory locations or 1,048,576 bytes
- Number of instructions increased.
    - from 45 on the 4004, to 246 on the 8085
    - over 20,000 variations on the 8086 & 8088

The 80286 Microprocessor

- Even the 1M-byte memory system proved limiting for databases and other applications.
  - Intel introduced the 80286 in 1983
  - an updated 8086
- Almost identical to the 8086/8088.
  - addressed 16M-byte memory system instead of a 1M-byte system
- Instruction set almost identical except for a few additional instructions.
  - managed the extra 15M bytes of memory

The 32-Bit Microprocessor

- Applications demanded faster microprocessor speeds, more memory, and wider data paths.
- Led to the 80386 in 1986 by Intel.
- Intel’s first practical microprocessor to contain a 32-bit data bus and 32-bit memory address.
- Through 32-bit buses, 80386 addressed up to 4G bytes of memory.
  - **1G** memory = 1024M, or 1,073,741,824 locations
- In 1989 Intel released the 80486, highly integrated package.
- 8K-byte cache memory system.

The Pentium Microprocessor

- Introduced 1993, Pentium was similar to 80386 and 80486 microprocessors.
- Originally labeled the P5 or 80586.
  - Intel decided not to use a number because it appeared to be impossible to copyright a number
- Introductory versions operated with a clocking frequency of 60 MHz & 66 MHz, and a speed of 110 MIPs.
- ***Pentium 2,3,4, Xeon and Core2, 64-bit and Multiple Core Microprocessors***

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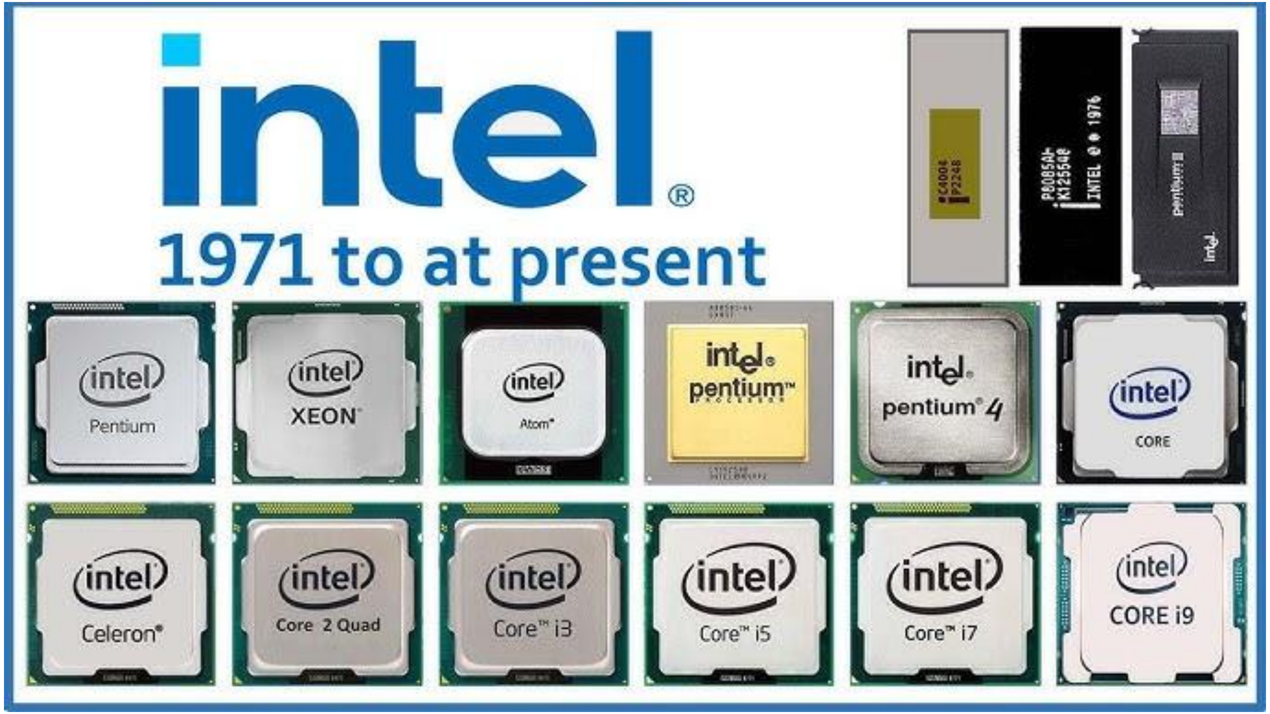
Intel Processors



Intel Processors

Manufacturer	Part Number	Data Bus Width	Memory Size
Intel	8048	8	2K internal
	8051	8	8K internal
	8085A	8	64K
	8086	16	1M
	8088	8	1M
	8096	16	8K internal
	80186	16	1M
	80188	8	1M
	80251	8	16K internal
	80286	16	16M
	80386EX	16	64M
	80386DX	32	4G
	80386SL	16	32M
	80386SLC	16	32M + 8K cache
	80386SX	16	16M
	80486DX/DX2	32	4G + 8K cache
	80486SX	32	4G + 8K cache
	80486DX4	32	4G + 16 cache
	Pentium	64	4G + 16K cache
	Pentium OverDrive	32	4G + 16K cache
	Pentium Pro	64	64G + 16K L1 cache + 256K L2 cache

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